

D3: Direction-Robust Sensor Count and Layout

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Does a layout survive an unknown heading or a directional sea?

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SHARC buoy project Code: `runDirectionRobustLayoutStudy.m`, `d3Tools.m`, `directionalSpectralMatrix.m`

Results: `Results/D3/D3_20261005_155648` (commit e2b057e)

Summary

Question. D1 found that two sensors suffice, but it assumed one *known* wave heading and every optimum lay on the incidence axis. D3 removes that assumption in two separate ways. **D3a:** one heading, but unknown; the model is still correct and the heading ψ is a nuisance parameter (Schur complement). **D3b:** a genuinely directional sea; the frozen single-direction model is wrong (V2), and each layout is judged on its own directional bias and covariance. Exhaustive over all 528, 5456 and 40 920 layouts of 2, 3 and 4 sensors, 24 headings, best reference per condition. **Outcome. An unknown heading raises the minimum count from two to three.** The best two-sensor layout is only Marginal (worst-case $\sigma_{\ln \beta} = 2.80\%$); three angularly spread sensors are Identifiable at every heading (0.60%), and a fourth still cuts that by 28% (0.43%), passing the 20% rule. The same holds at all three noise levels. The cost is geometry, not the heading estimate: the robust layouts lose only 0.2 to 2.4% to estimating ψ , whereas the D1 axis layouts keep only 3 to 4% of the robust worst-case efficiency at $n_s = 3, 4$. **In directional seas no layout with up to four sensors passes the $d_{\text{tot}} \leq 0.5$ gate, in any single registered sea.** Placement reduces the two-component bias by one to two orders of magnitude (to $d_{\text{tot}} \approx 1$ at $n_s = 4$), but the directional-spread sea stays at $d_{\text{tot}} \approx 25$ to 33.

1 What the study does

Why D3 exists. D1 forced the floe from one known heading, so the transmissibility varied most along the incidence axis and every optimum lay on it. In the field the heading is estimated, not known (D3a), and the sea has spread or several components, which V2 showed breaks the single-direction model (D3b). The two problems are kept apart: in D3a the model is right and only information is lost; in D3b the model is wrong and the question is whether a layout can make the error small enough. Design otherwise as D1: the 33-point candidate grid, p_0 , setting F, 0.10 m spacing, LSM6DSV16X headline with ICM-42688-P and G3 repeated, reference optimised (a processing choice, made per condition).

2 Method

Pre-registered in programme rev. 4 and `cfg.D3`; clarifications and one amendment logged before the run; tester 14/14.

Headings by symmetry. A heading ψ samples the nominal field at points rotated by $-\psi$. The grid repeats every 45° , so heading $45k + \delta$ of a layout equals heading δ of that layout rotated by $-45k$. One EMM field at the D1 grid and its copies at $\delta = 7.5, 22.5$ and 37.5° (plus $\pm 0.5^\circ$ copies for $\partial T / \partial \psi$) gives all 24 headings by index permutation; the tester confirms this against a copy sampled directly at 52.5° (7×10^{-13}).

Heading grid (amendment). The registered headings $0, 15, \dots, 345^\circ$ were replaced by the midpoint grid $7.5 + 15k^\circ$ before the run. On a grid axis a layout whose sensors all lie on that diameter (Level 2C, the D1 optimum) has $\partial T / \partial \psi = 0$ by mirror symmetry, so $F_{\psi\psi} = F_{\theta\psi} = 0$ and the Schur complement is 0/0 at a measure-zero heading. The midpoint grid is the same uniform quadrature without it. Heading 0 is kept, with the heading known, as the D1 gate: it reproduces D1's two-sensor optimum (s6 + s18, reference s18).

D3a: unknown heading as a nuisance. With $\vartheta = [\ln \beta, \ln R, \psi]$ and F partitioned into the $\theta = [\ln \beta, \ln R]$ and ψ blocks,

$$F_{\text{eff}} = F_{\theta\theta} - F_{\theta\psi} F_{\psi\psi}^{-1} F_{\psi\theta}, \quad A(\psi) = \text{tr } F_{\text{eff}}^{-1}.$$

Scored *per heading*, then $\bar{A}$ (mean) and A_{worst} (max) over the 24 headings; a heading-averaged F is never inverted, since that would describe an experiment repeated at every heading. The penalty is $\sigma_{\ln \beta}(\psi \text{ estimated}) / \sigma_{\ln \beta}(\psi \text{ known})$.

D3b: directional seas. Second component at $\phi = 30, 60, 90^\circ$ with $f_2 = 10$ and 25%, and one $\cos^{2s}$ spread ($s = 10$, 24 directions), each rotated with the primary heading over all 24 headings; primary heading known (isolating the model error from D3a), headline noise. For every layout, sea and heading: Σ from the rank-two (or spread) spectral matrix, F and projection weights under that Σ , $b_{\text{dir}} = f_W^{\text{dir}} - f_W^{\text{single}}$, d_{tot} from $\delta\theta_W + \delta\theta_{\text{dir}}$. **Gate:** a layout is feasible only if $d_{\text{tot}} \leq 0.5$ in every sea and heading, using for each condition the reference that passes with the smallest A . Choosing the reference with knowledge of the bias is optimistic, so a negative result is robust. The gate is never relaxed; the least-bad layout (smallest worst-case d_{tot}) is a diagnostic.

Count rule (as D1, on the worst-case optimum): a sensor is worth adding if it lowers the worst-case $\sigma_{\ln \beta}$ by $\geq 20\%$; the recommended count is the smallest n_s whose worst-case optimum is Identifiable at every heading, and with D3b also bias-feasible.

3 Results: D3a, unknown heading

n_s	Design	Layout	worst-case $\sigma_{\ln \beta}$		mean $\sigma_{\ln \beta}$	worst class	efficiency mean / worst
			ψ estimated	ψ known			
2	D3a robust	s2 + s30	2.80%	2.60%	1.29%	Marginal	1.00 / 1.00
2	D1 optimum	s6 + s18	5.18%	5.18%	1.59%	Not identifiable	0.52 / 0.29
3	D3a robust	s18 + s20 + s23	0.60%	0.59%	0.50%	Identifiable	1.00 / 1.00
3	D1 optimum	s6 + s14 + s18	3.40%	3.06%	1.15%	Marginal	0.14 / 0.03
4	D3a robust	s18 + s20 + s22 + s24	0.43%	0.43%	0.40%	Identifiable	0.99 / 1.00
4	D3a mean-A	s10 + s20 + s24 + s30	0.48%	0.48%	0.41%	Identifiable	1.00 / 0.81
4	D1 optimum	s6 + s14 + s18 + s26	2.17%	2.16%	0.86%	Marginal	0.18 / 0.04
4	Level 2C	s2 + s6 + s14 + s26	1.45%	1.45%	0.76%	Marginal	0.26 / 0.09

LSM6DSV16X. Worst/mean over the 24 headings, best reference per heading. Efficiency: the robust optimum's $\bar{A}$ (or A_{worst}) over the design's. At $n_s = 2, 3$ the mean-A and worst-case optima are the same layout up to a 45° rotation or mirror (identical metrics). Layouts Identifiable at every heading: 0 of 528, 740 of 5456, 21 760 of 40 920.

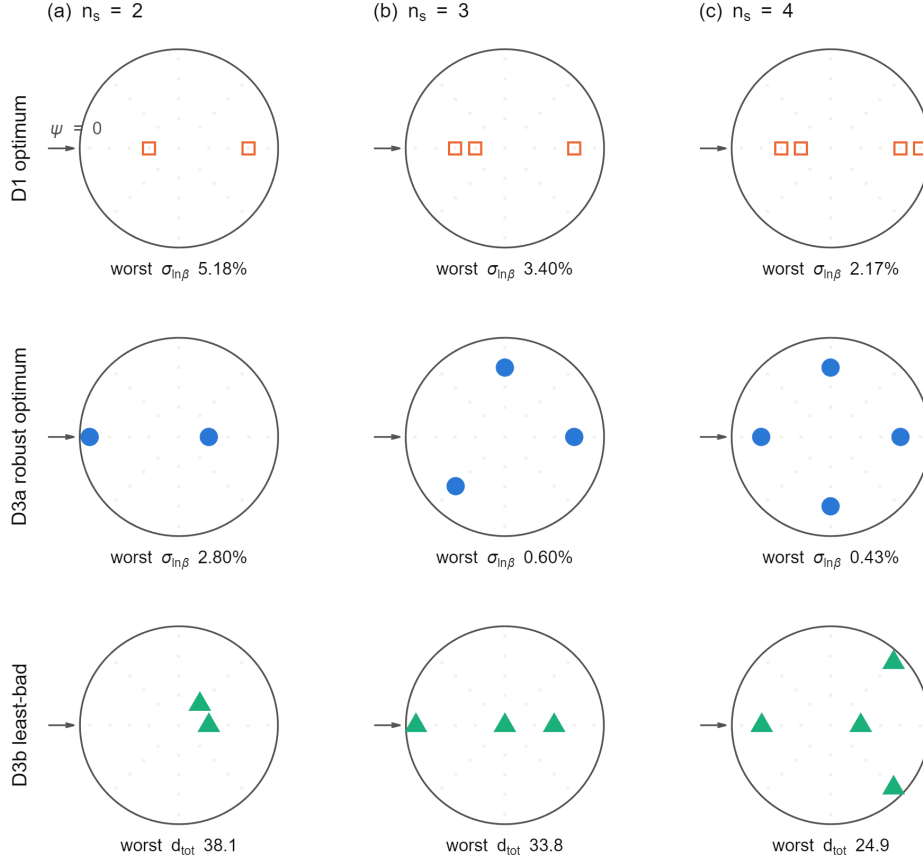


Figure 1. Selected layouts (rows) for $n_s = 2, 3, 4$ (columns): the D1 optimum (heading 0 known), the D3a robust (worst-case) optimum (heading unknown) and the D3b least-bad layout (smallest worst-case d_{tot} over all seas and headings). Grey: the candidate grid; arrow: heading $\psi = 0$.

Two sensors cannot be robust. Any two points define a line, and some heading is nearly perpendicular to it; even the best pair (s2 + s30, a diameter at 0.3R and 0.9R) is only Marginal. A third sensor off the line removes that blind direction: the robust three- and four-sensor layouts are spread around the $0.7R$ ring (three at 0, 90 and 225° ; the four-sensor optimum is a symmetric cross).

Noise	worst-case optimum, worst $\sigma_{\ln \beta}$			gain		recommended n_s
	$n_s = 2$	3	4	2 $\rightarrow$ 3	3 $\rightarrow$ 4	
LSM6DSV16X	2.80%	0.60%	0.43%	79%	28%	3
ICM-42688-P	3.87%	0.82%	0.59%	79%	28%	3
G3 assumption	3.73%	0.79%	0.57%	79%	28%	3

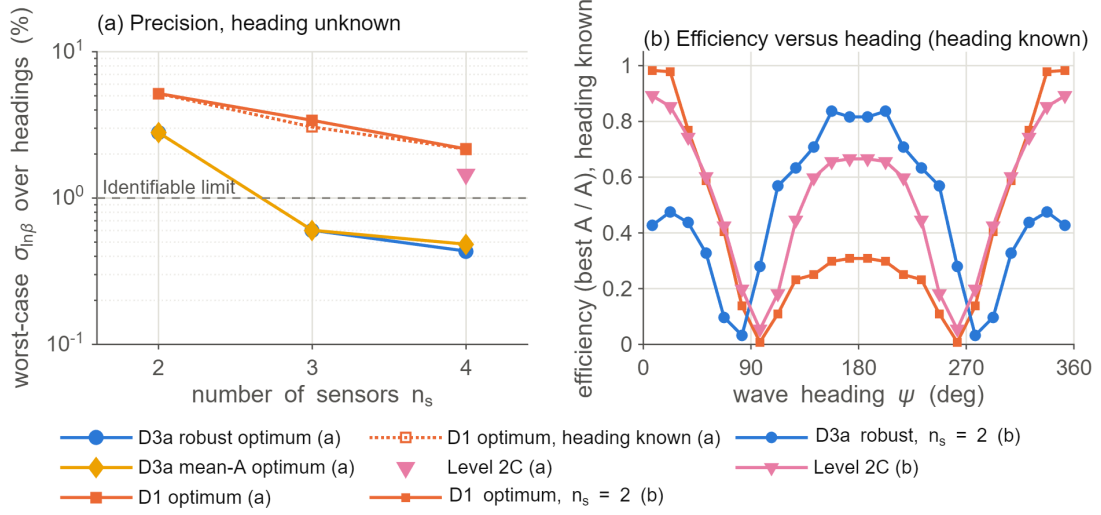


Figure 2. (a) Worst-case $\sigma_{\ln \beta}$ over the headings (heading estimated) against n_s for the D3a optima, the D1 optimum (also with the heading known) and Level 2C, with the 1% Identifiable limit. (b) Efficiency against heading with the heading known: best A at that heading over the design's A .

Geometry, not the heading estimate, is the cost. For the robust layouts the nuisance penalty is small: 0.602% against 0.588% at $n_s = 3$ (+2.4%) and 0.434% against 0.433% at $n_s = 4$. The D1 layouts fail because they are lines along one diameter: at the wrong heading their efficiency falls towards zero (Figure 2b), and their worst-case $\sigma_{\ln \beta}$ is 5 to 6 times the robust optimum's at $n_s = 3, 4$. For two-sensor layouts the penalty reaches about 1.9 at individual headings (Figure 3a), but there the layout itself is the limit.

4 Results: D3b, directional seas

No layout passes the bias gate: 0 of 528, 5456 and 40 920 at $n_s = 2, 3, 4$, and 0 in *every individual* sea, so the result is not an artefact of requiring all seas at once.

Sea	least-bad worst-case d_{tot} over 24 headings (precision-optimal reference)		
	$n_s = 2$	$n_s = 3$	$n_s = 4$
$\phi = 30^\circ, f_2 = 10\%$	17.1 (17.5)	2.92 (6.88)	1.00 (1.46)
$\phi = 30^\circ, f_2 = 25\%$	33.5 (35.6)	3.64 (8.70)	0.97 (1.28)
$\phi = 60^\circ, f_2 = 10\%$	20.1 (21.2)	2.83 (4.05)	0.99 (1.33)
$\phi = 60^\circ, f_2 = 25\%$	37.5 (38.1)	3.06 (5.12)	1.23 (1.75)
$\phi = 90^\circ, f_2 = 10\%$	20.1 (22.6)	2.43 (4.60)	1.19 (2.35)
$\phi = 90^\circ, f_2 = 25\%$	35.0 (39.6)	2.96 (6.11)	1.34 (2.82)
$\cos^2 s$ spread, $s = 10$	28.6 (29.3)	32.6 (37.2)	24.9 (33.4)
All seas (one layout)	38.1 (s2 + s3)	33.8 (s1 + s10 + s30)	24.9 (s2 + s22 + s27 + s33)

Per sea: the minimum over layouts (a different layout may achieve each). Gate $d_{\text{tot}} \leq 0.5$; feasible layouts: 0 in every cell. In brackets: with the precision-optimal reference instead of the bias-aware one.

Placement helps a great deal against discrete components. For the two-component seas the least-bad d_{tot} falls from 17 to 38 at $n_s = 2$, to 2.4 to 3.6 at $n_s = 3$ and to 0.97 to 1.34 at $n_s = 4$: one to two orders of magnitude, yet still above the gate. The bias-aware reference matters (at $90^\circ, 25\%$: 1.34 against 2.82), and even this optimistic freedom does not reach 0.5. **The spread sea is not rescued:** 28.6, 32.6, 24.9. It does not fall with n_s ; the rise at $n_s = 3$ is possible because d_{tot} is measured in units of a precision that improves with each sensor, so a bias that does not shrink becomes more significant. With every heading component present, no sensor geometry separates the directions.

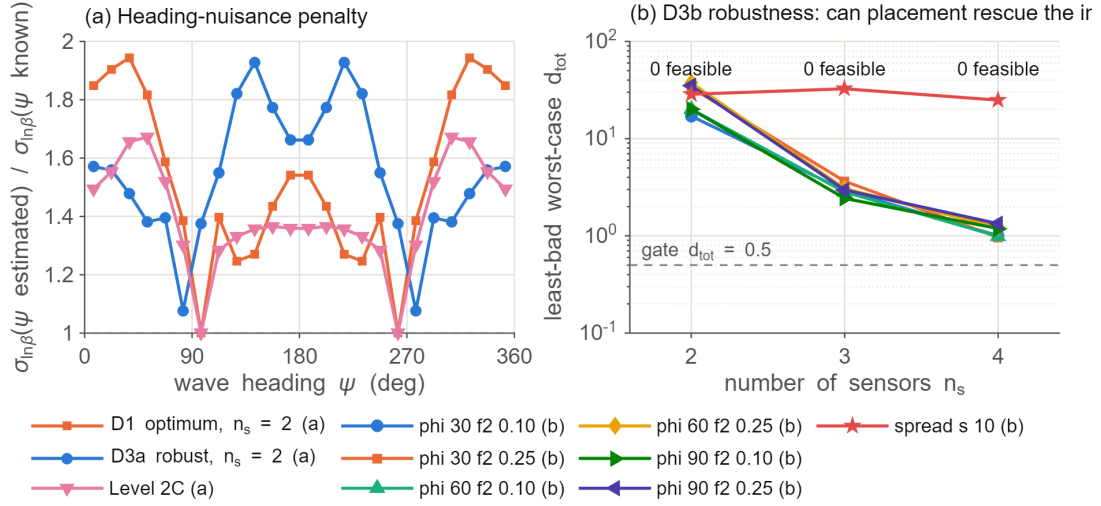


Figure 3. (a) Heading-nuisance penalty $\sigma_{\ln\beta}(\psi \text{ estimated})/\sigma_{\ln\beta}(\psi \text{ known})$ against heading for two-sensor designs and Level 2C. (b) D3b: least-bad worst-case d_{tot} per sea against n_s , with the $d_{\text{tot}} = 0.5$ gate; no layout is feasible at any n_s .

5 Interpretation

1. **D1's two-sensor result is conditional on a known heading.** With the heading unknown, three angularly spread sensors are the minimum Identifiable configuration, at every noise level tested.
2. **Estimating the heading is cheap; placing for the wrong heading is not.** Once the geometry is robust the Schur penalty is 0.2 to 2.4%; at $n_s = 3, 4$ the D1 and Level 2C axis layouts keep only 3 to 9% of the robust worst-case efficiency.
3. **A fourth sensor is still worthwhile:** it is not needed for identifiability but cuts worst-case $\sigma_{\ln\beta}$ by 28%, above the 20% rule.
4. **Placement mitigates discrete multidirectional bias** by one to two orders of magnitude between two and four sensors, to $d_{\text{tot}} \approx 1$.
5. **Placement cannot fix a structurally wrong estimator.** No layout passes the gate in any registered sea, and the spread sea stays at $d_{\text{tot}} \approx 25$ to 33 regardless of n_s .

Conclusion. D1 showed that two sensors suffice when the heading is known. With a single but unknown heading, two sensors on any line are only Marginal, and three angularly distributed sensors are the minimum Identifiable design (worst-case $\sigma_{\ln\beta} = 0.60\%$ at LSM6DSV16X noise); a fourth remains worthwhile, reducing it by 28%. The penalty for estimating the heading is small once the geometry is robust, so the dominant loss is placing sensors along a single assumed axis. Under genuinely directional seas, sensor count and placement reduce the bias of the frozen single-direction inverse by one to two orders of magnitude for discrete secondary components, but no layout with up to four sensors satisfies the registered $d_{\text{tot}} \leq 0.5$ criterion, and a spread sea remains strongly inconsistent with the model. The limitation is structural: directional forcing must be detected and excluded or represented in the inference model; it is not a sensor-placement problem.

Limitations. 33 discrete candidate points and $n_s \leq 4$; one floe (p_0). D3a samples 24 headings; the heading derivative is a $\pm 0.5^\circ$ central difference (stable to 0.06% when halved). D3b at the headline noise only, with the primary heading known (D3a and D3b effects are not combined), seven registered seas (one spread, $s = 10$). The bias-aware reference choice is optimistic. d_{tot} is a precision-scaled measure: the least-bad layouts' parameter biases ($\delta\beta/\beta$) were not stored, so the spread result is stated in Fisher units. Apart from the forcing, truth and inverse share the EMM.

Requirement for the experiment and the thesis

For a single but unknown incident heading, use at least three angularly distributed sensors (the robust layouts are spread around the $0.7R$ ring); four remain worthwhile where practical. Do not use the D1 incidence-axis two-sensor design (or Level 2C) unless the heading is known. For genuinely multidirectional seas, no tested layout with up to four sensors satisfies the registered $d_{\text{tot}} \leq 0.5$ consistency criterion: placement reduces directional bias substantially but cannot guarantee validity of the frozen single-source inverse. Directional forcing must therefore be detected and excluded (V2: coherence, with a calibrated noise floor) or incorporated explicitly into the inference model. The Conclusion above is the thesis wording.

Outputs (Results/D3/): D3_20261005.155648.log/.mat/_summary.txt; D3_d3a.csv, D3_d3b.csv, D3_counts.csv; figures/D3_fig1 to D3_fig3 (.pdf/.png) from plotDirectionRobustLayoutStudy. Tester testDirectionRobustLayoutStudy 14/14; run 1906 s.